DEVELOPMENT OF AN INTEGRATED HOSPITAL INFORMATION SYSTEM FOR DR. FRANCISCO C. PARCON CLINIC AND HOSPITAL

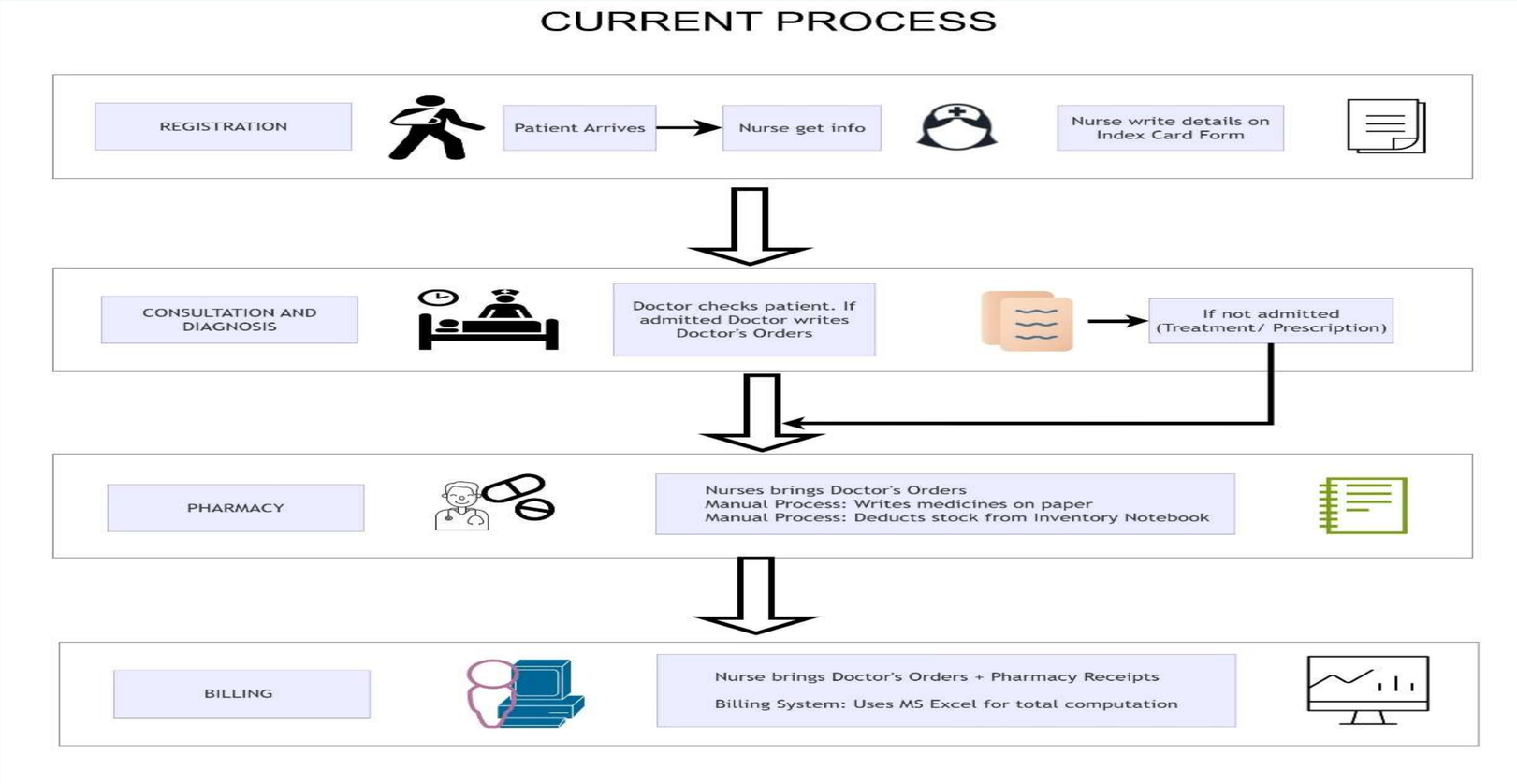
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BAGONG PILIPINAS

TÜVRheinland CERTIFIED
Management System
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- The Dr. Francisco C. Parcon Clinic and Hospital was established in 2017 to provide basic healthcare services to the residents of Barangay Rizal and the wider municipality of Banga, South Cotabato.
- Dr. Francisco C. Parcon Clinic and Hospital has expanded from a local clinic to a fully accredited hospital, managing 18 admissions and over 25 check-ups daily.
- The current paper-dependent workflow is slow, inefficient, and requires physical "manual walking" of documents between departments.

- When a patient arrives, the nurse writes the patient's information on an index card.
- Doctors write their orders by hand, while medicine transactions and inventory deductions are recorded manually in a notebook.
- Billing staff then collect the necessary information and use MS Excel to calculate the patient's total bill.
- Information has to be passed manually from one department to another.



PROBLEM STATEMENT

- Nurses have to physically carry prescriptions and charge slips from one department to another.
- Patient information is repeatedly recorded in different areas, increasing the risk of errors, inconsistencies and mismatched records.
- In the pharmacy, inventory is tracked manually.
- Billing Department has to wait for physical slips to arrive, which often leads to delayed patient discharge.

GENERAL OBJECTIVES

Generally, this project is intended to design and develop an Integrated Hospital Information System for Dr. Francisco C. Parcon Clinic and Hospital.

SPECIFIC OBJECTIVES

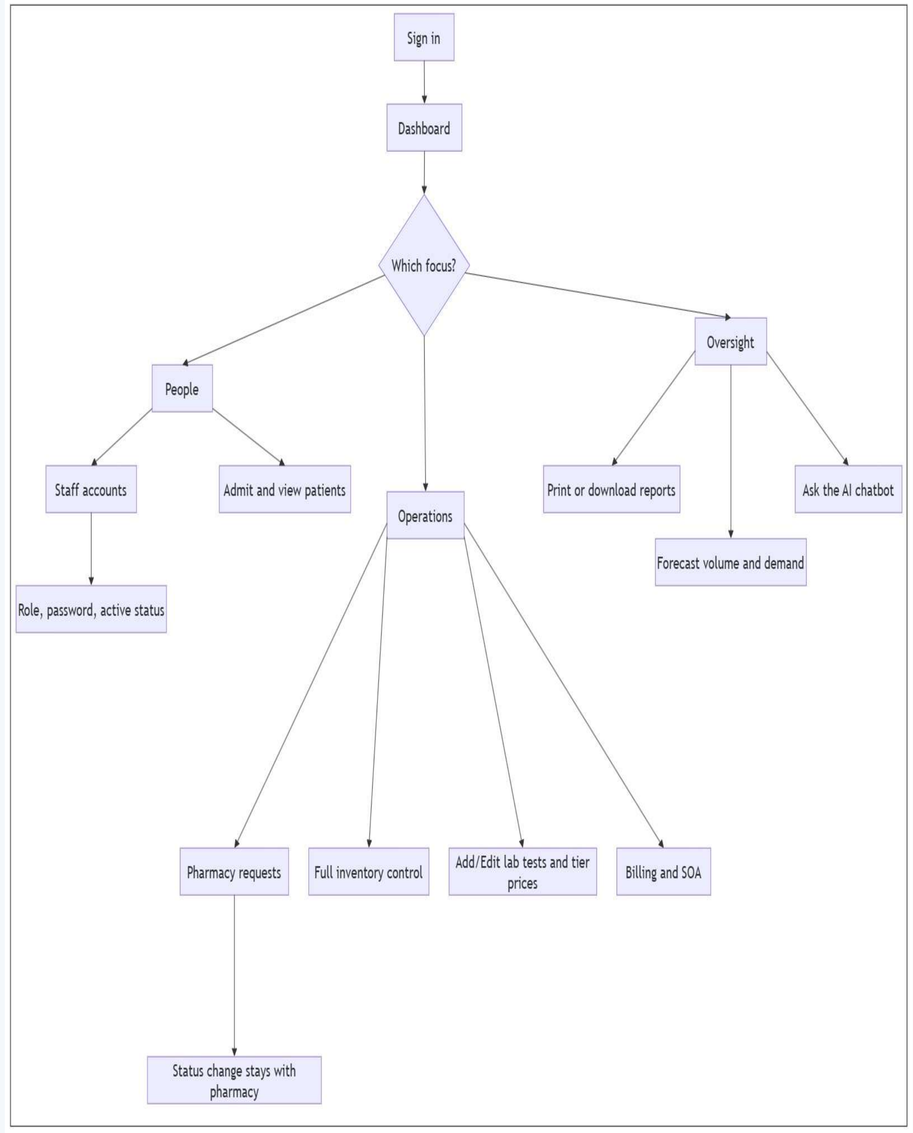
- develop a user management module that enables the admin to manage user accounts, patient records, pharmacy request oversight, medicine and laboratory pricing, billing, and hospital reports and forecasts;
- develop a nursing management module that allows nurses to manage patient admissions and check-ins, record patient information and vital signs, coordinate pharmacy and laboratory requests, and monitor patient status;
- develop a pharmacy management module that enables pharmacy staff to process medicine requests and counter sales, manage stock levels, and monitor inventory through an AI-powered chatbot;
- develop an automated billing module that captures hospital charges, records payments, and generates itemized billing statements; and
- evaluate the usability of the developed system in terms of ease of use for admin, nurses, pharmacists, and billing staff.

EXPECTED OUTPUTS

- A user management module for the admin to create and manage user accounts and oversee hospital operations across the web dashboard.
- A mobile-responsive nursing management system, limited to the nurse portal and nurse mobile login.
- A pharmacy dispensing system with an AI-powered inventory chatbot accessible to pharmacy staff through the web dashboard.
- An automated billing system for billing staff.
- An easy-to-use system that supports all user roles – admin, nurses, pharmacy staff, and billing staff.

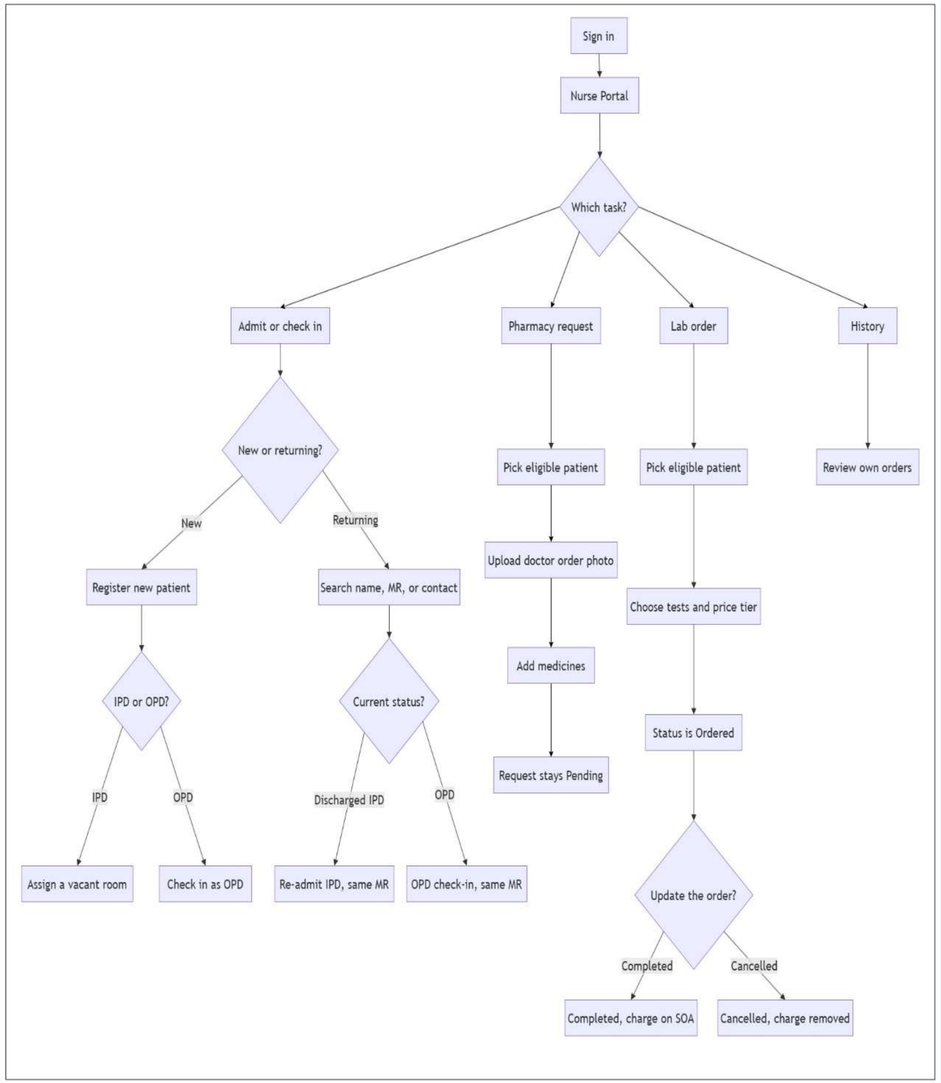
PROPOSED PROCESS FOR ADMIN

- The admin can access everything through a single dashboard.
- The admin can manage user accounts, admit and view patients, check pharmacy requests, and view hospital reports and forecasts.
- The admin can add, edit, delete, and restock medicines, and print or download the inventory list.
- The admin can also add, edit, and delete lab tests and their prices, create and edit bills, record payments, and print or download the Statement of Account.



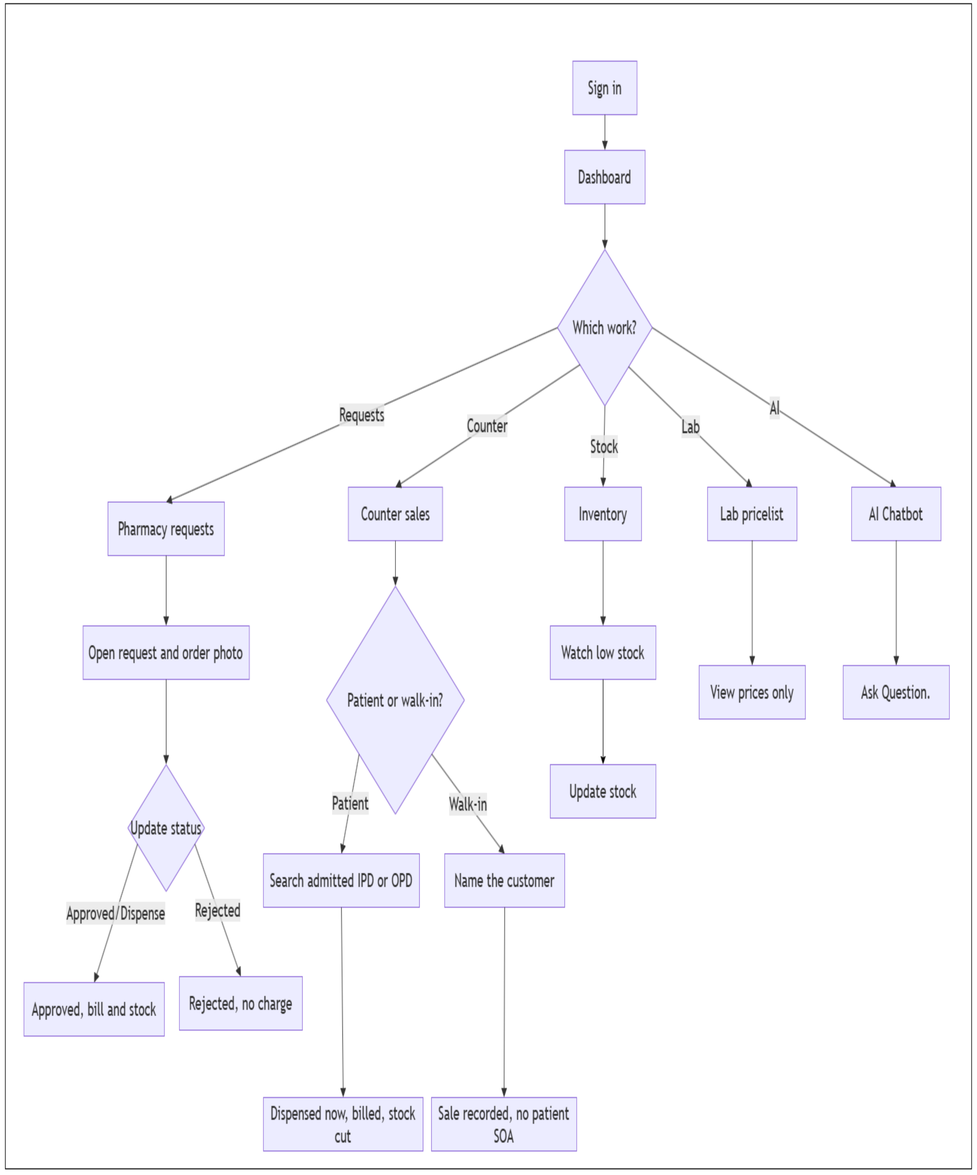
PROPOSED PROCESS FOR NURSE

- Nurses can register or admit a new patient, search for an existing patient, send pharmacy requests with a photo of the doctor's order, request laboratory tests based on the available price category, and check the status of submitted requests.
- A nurse records patient details and vital signs, then updates or discharges only the inpatients they admitted.
- This helps reduce the need to manually carry papers between departments and makes communication faster and more organized.



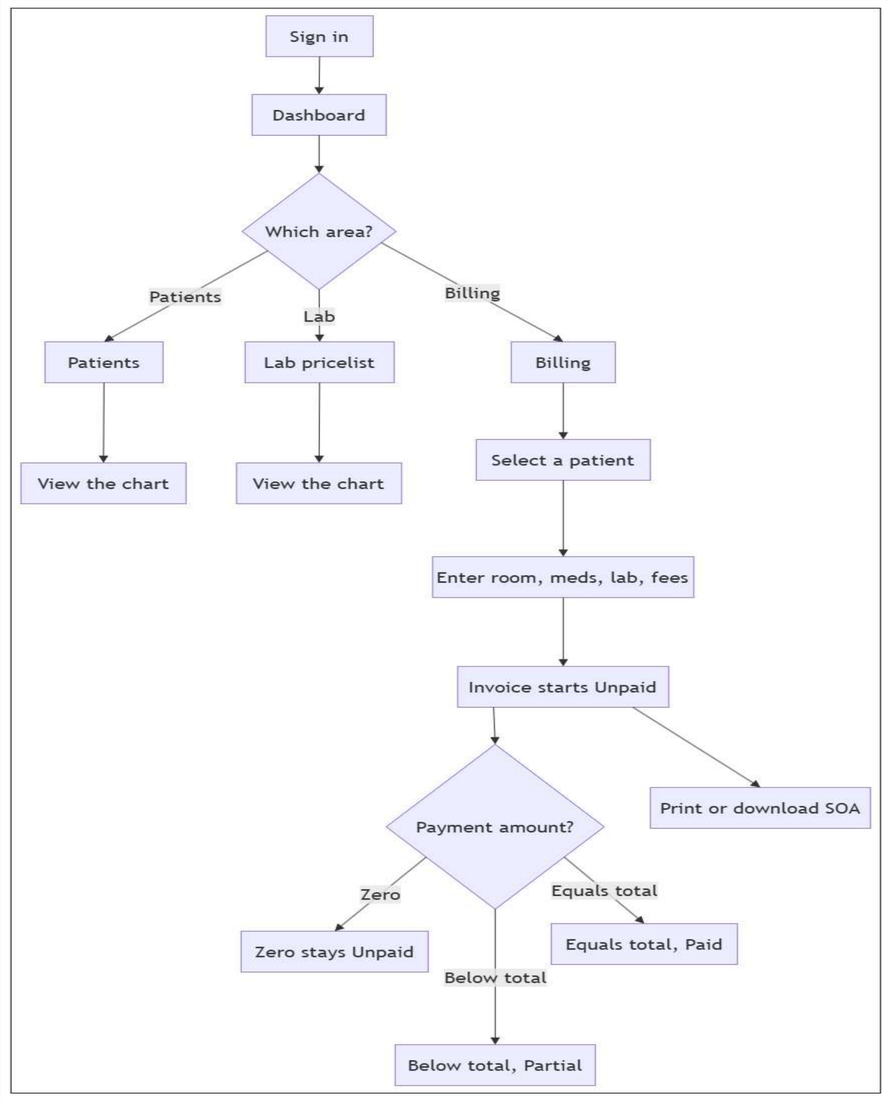
PROPOSED PROCESS FOR PHARMACY

- The pharmacist can view incoming medicine requests, check the attached doctor's order, and update the request status directly in the system.
- The pharmacist can also process medicine sales for admitted patients and walk-in customers, manage medicine stocks, check medicines that are running low, and use the AI chatbot to ask about medicine availability.
- Approving a request adds the medicine charge to the patient's bill, and dispensing reduces stock.

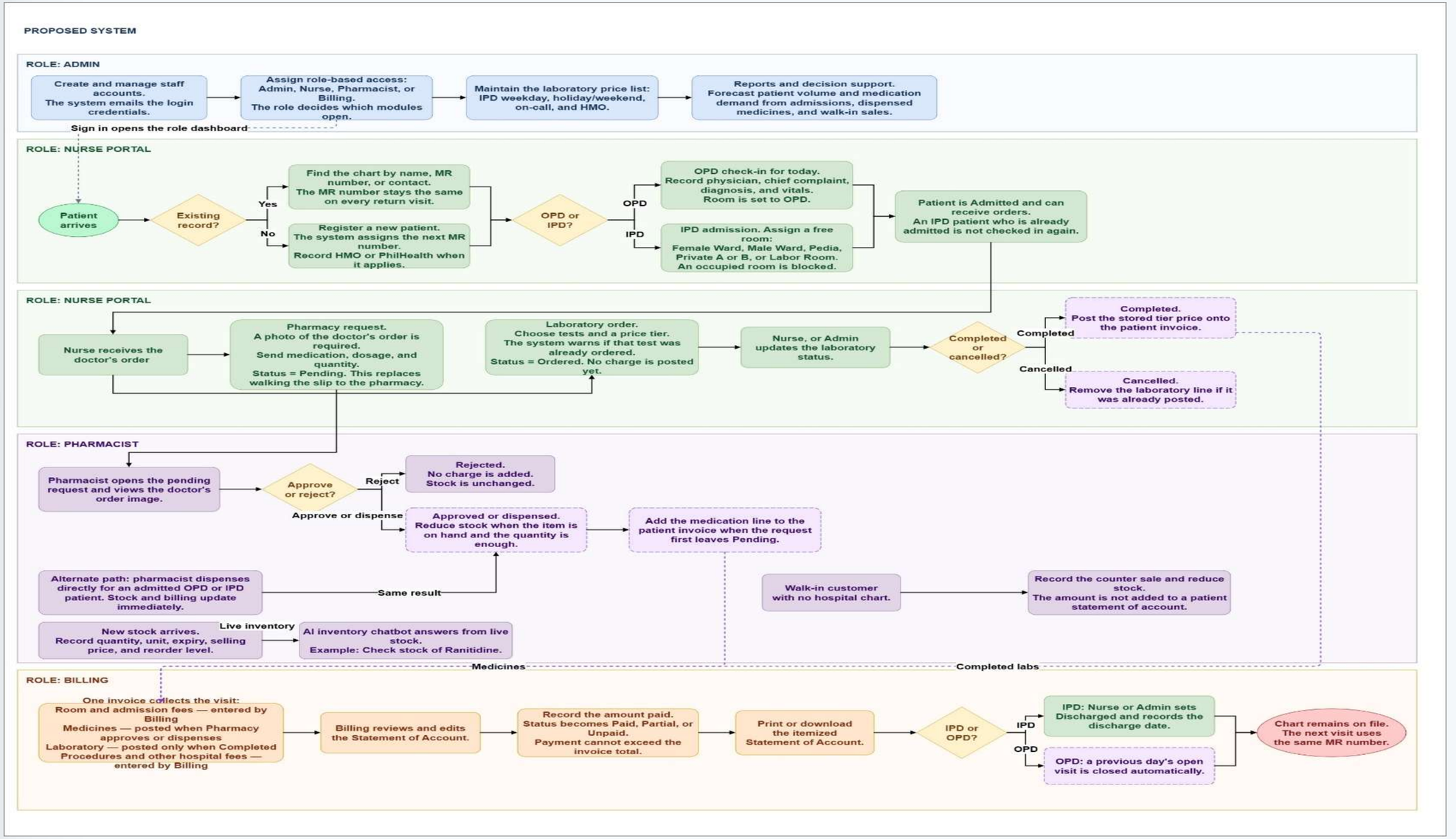


PROPOSED PROCESS FOR BILLING

- The billing staff can view the patient's charges in one place.
- The patient's invoice starts with an Unpaid status.
- Billing staff can select the patient, review the charges, record payments, and monitor the remaining balance.
- Once the account has been settled, they can print or download the patient's itemized Statement of Account.



PROPOSED SYSTEM



SCOPE AND LIMITATIONS

- The proposed project will focus on user management, nursing management, pharmacy management, and automated billing modules.
- The system is intended for internal hospital use and will not include modules for Laboratory or Radiology results.
- Laboratory requests within the system are limited to ordering and pricing for billing purposes only.
- It also will not feature an external website for patient appointments during this phase.
- It will not handle broader financial tasks like staff payroll, tax reporting, or full-scale institutional accounting.

SIGNIFICANCE OF THE STUDY

- **Hospital Staff:** Reduces paperwork and automates crucial functions like records, stock, and billing. It streamlines routines, helping employees work efficiently and communicate better.
- **The Hospital:** Enhances operations through a centralized, organized management of information, allowing the institution to deliver more accurate and high-quality health care.
- **Future Researchers:** This project can be a guide and reference for future researchers who want to develop or improve hospital information systems and other healthcare technologies

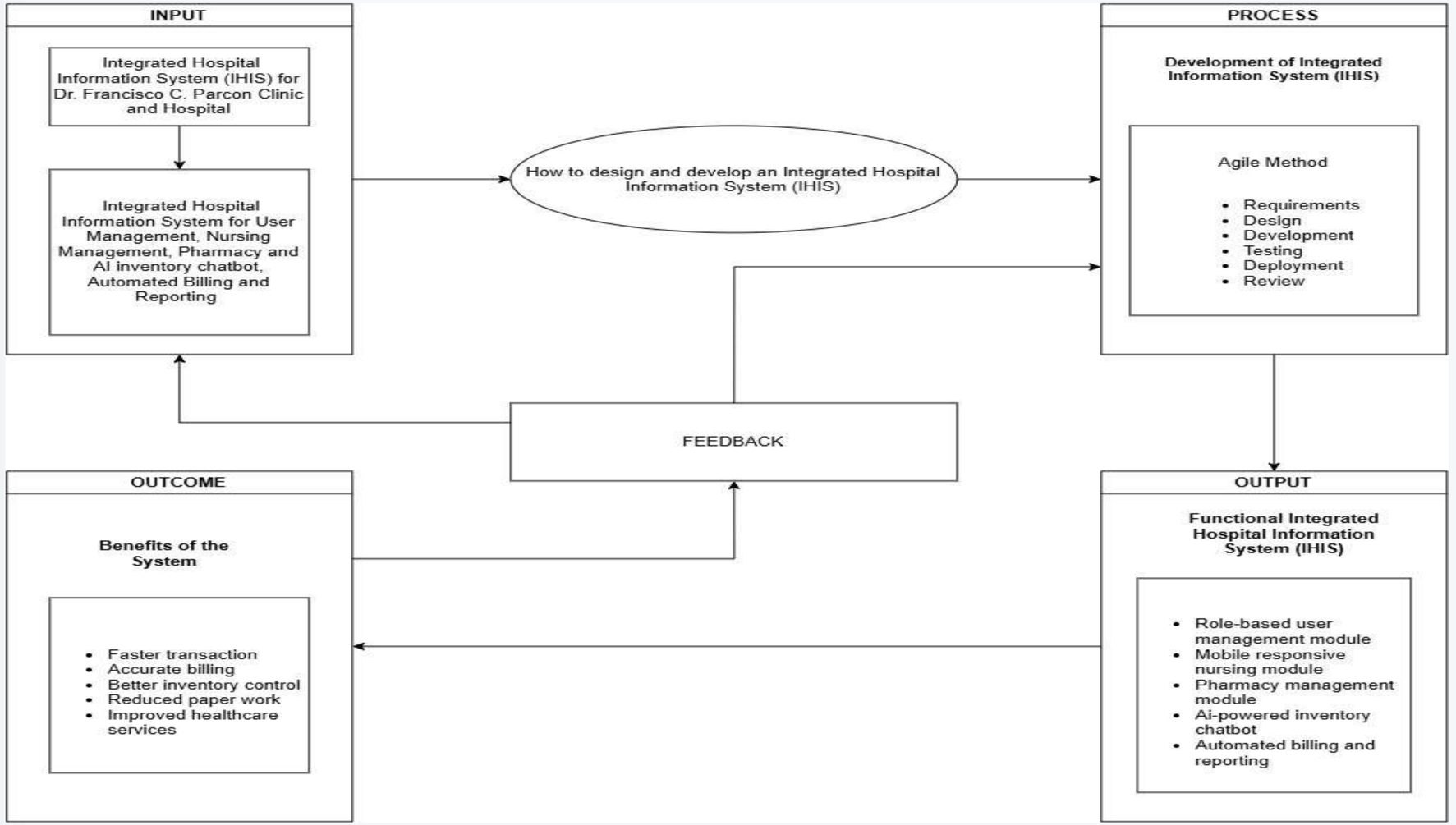
PEOPLE

- **Admin**
- Can view or edit every patient record, including discharge and visit history.
- Can view pharmacy requests and submit a request, but cannot approve or dispense it.
- **Nurse**
- Pharmacy and lab requests are limited to the nurse's own patients, and an outpatient request is allowed only for today's visit.
- Can view the laboratory pricelist but cannot modify prices, inventory, bills, user accounts, reports, or the chatbot.

PEOPLE

- **Pharmacy Staff**
- Can view the lab pricelist but cannot admit or edit patients, open billing, manage users, or open reports.
- **Billing Staff**
- Can view patient records and visit history but cannot admit a patient or change the clinical chart.
- Cannot open pharmacy, inventory, user management, reports, Decision Support, or the AI chatbot modules

CONCEPTUAL FRAMEWORK



SYSTEM OVERVIEW

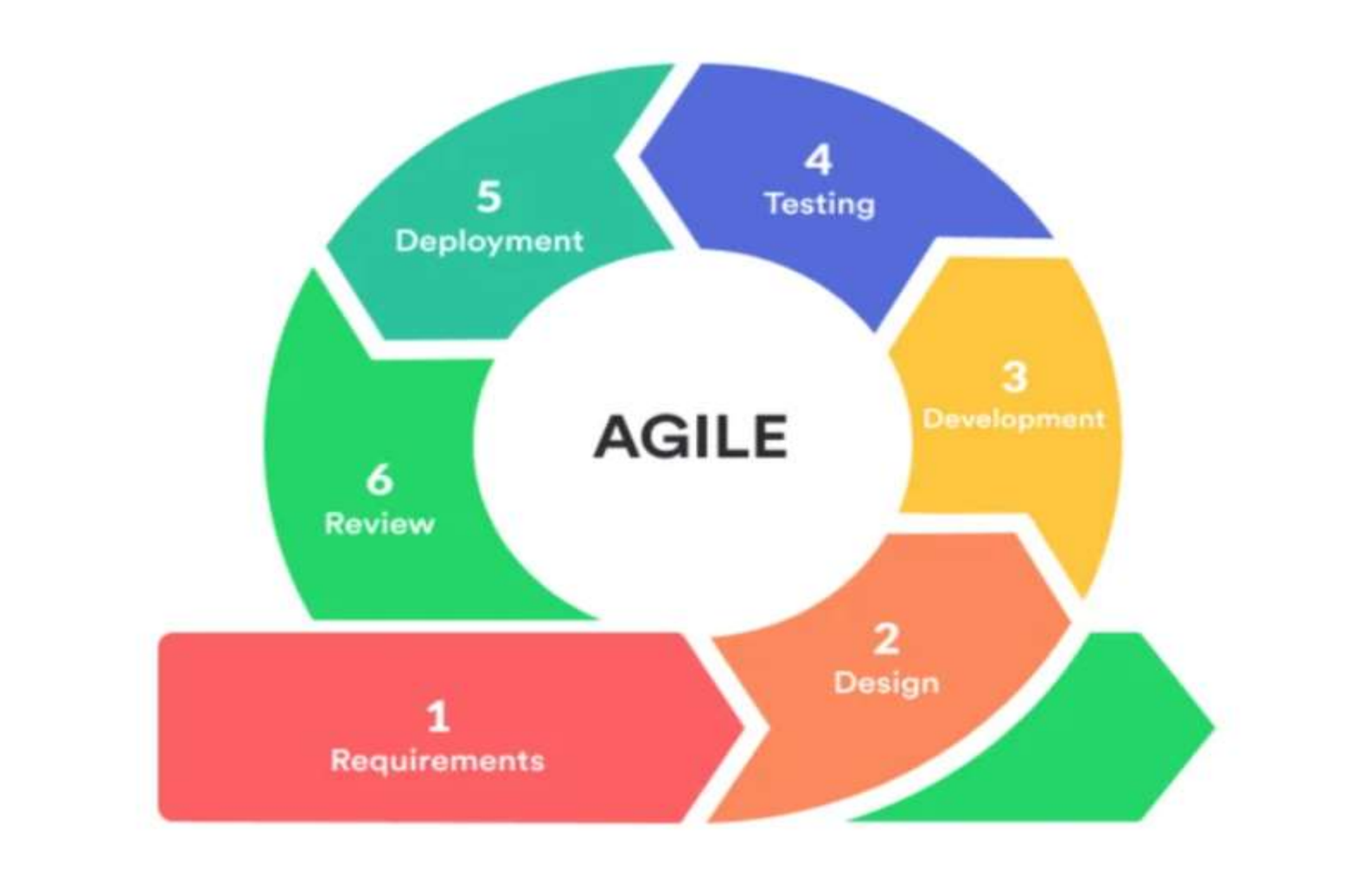
- **Data:** Securely stores user accounts, clinical records, pharmacy requests, lab orders, and itemized billing transactions.
- **Hardware:** Operates on standard Windows 10+ PCs (Core i5, 4-8GB RAM). Nurses access the portal via mobile devices.

METHODOLOGY

- **Research Design:** Uses a quantitative design with structured questionnaires to evaluate system functionality and user satisfaction.
- **Respondents:** 30 hospital staff (nurses, pharmacists, billing, admins) using 4-point Likert scale questionnaires.
- **Data Collection:** Gathers data via questionnaires after hands-on testing, securing formal approval and informed consent first.
- **Data Analysis:** Summarizes evaluation results using descriptive statistics and a 4-point Likert scale.
- **Validity:** Ensures all system requirements are met through functional testing and photographic documentation.
- **Ethical considerations:** Protects participant confidentiality and records all data objectively to maintain integrity.

PROJECT DEVELOPMENT APPROACH

AGILE METHODOLOGY

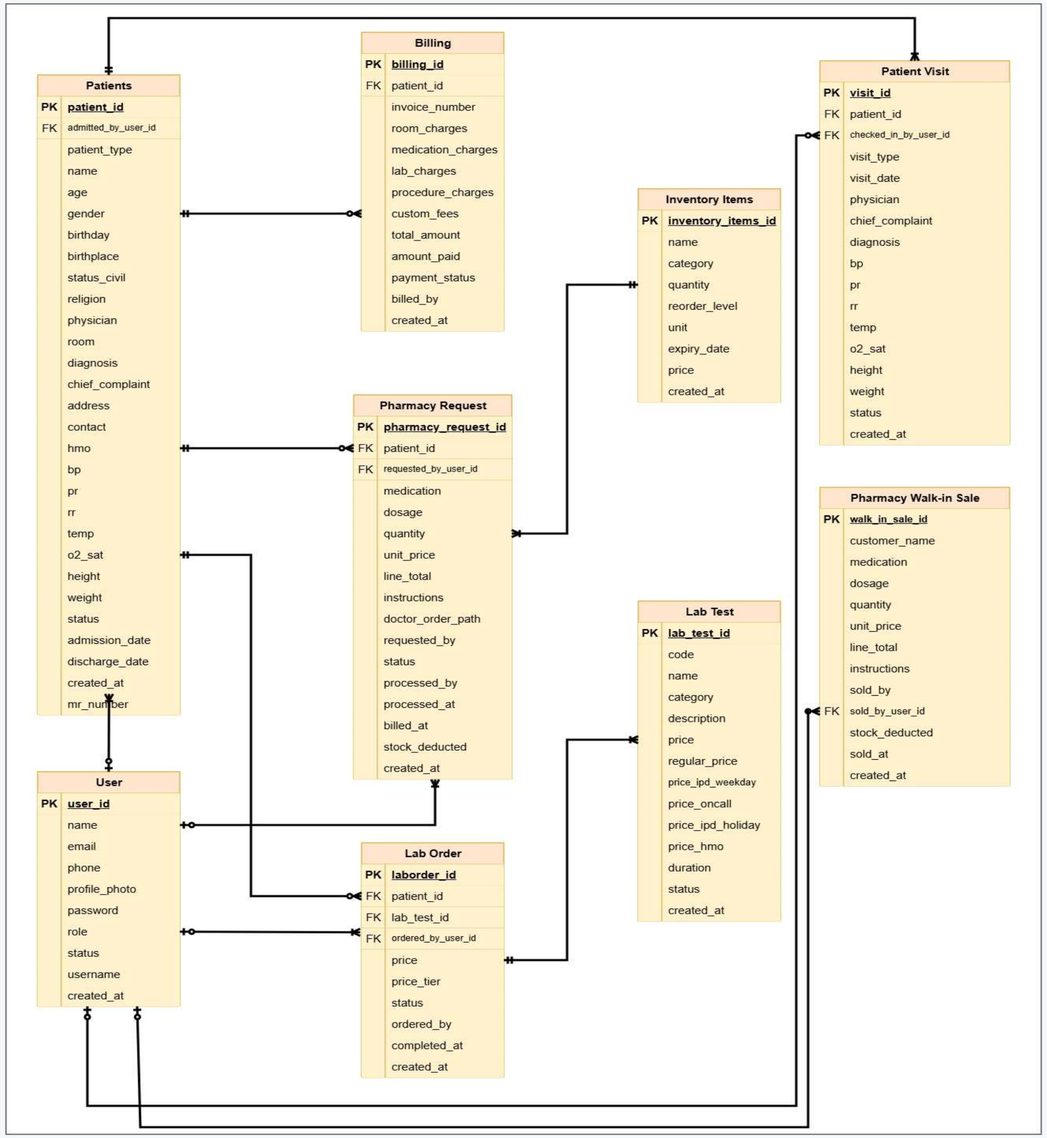


PROJECT DEVELOPMENT APPROACH

- **Planning:** Defining requirements, identifying hospital constraints, and mapping out user roles.
- **Design:** Creating UML diagrams, UI/UX wireframes, and database relationship structures.
- **Development:** Iterative coding of web dashboards and mobile portals with continuous feedback loops.
- **Testing:** Usability evaluation using Likert scales across all respondent groups.

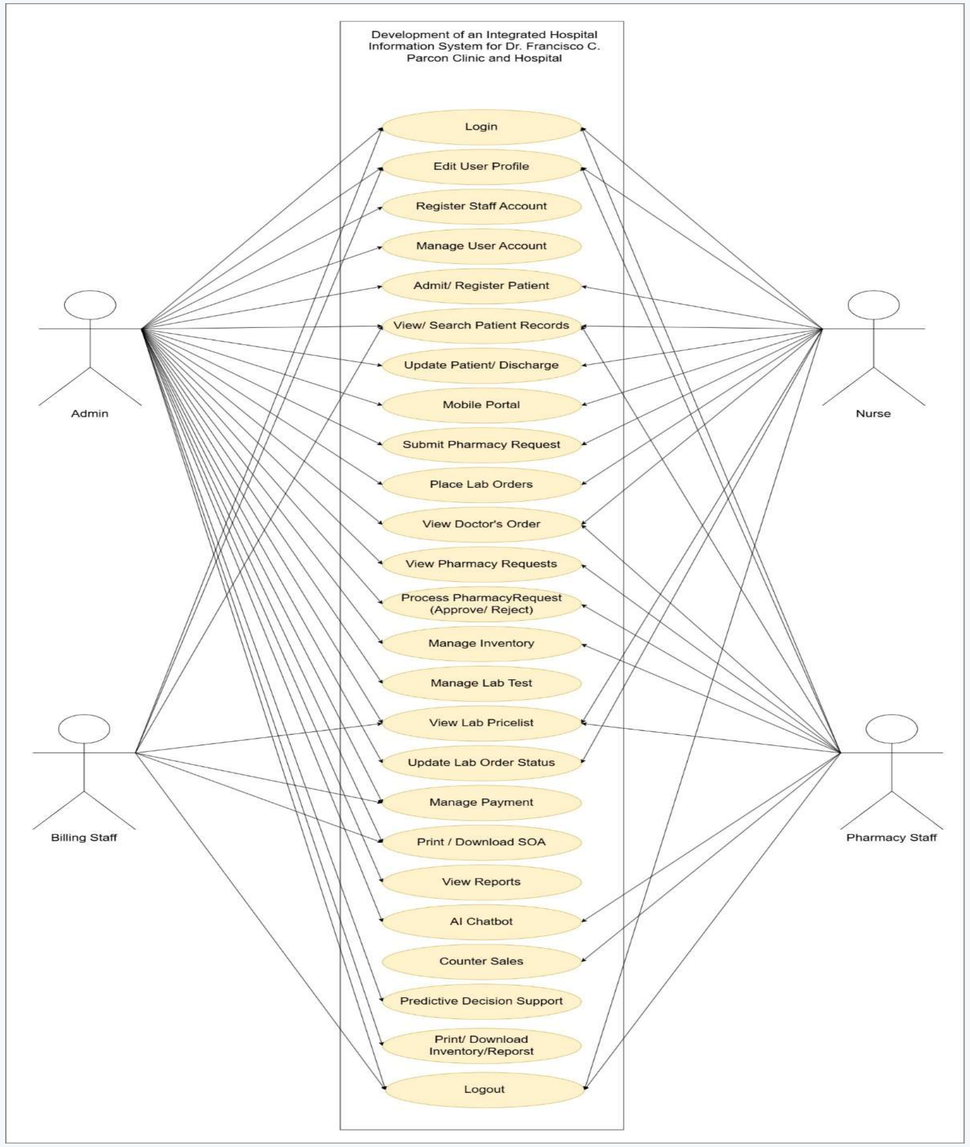
Entity Relationship Diagram (ERD)

- The main entities are User, Patients, Patient Visit, Pharmacy Request, Inventory Items, Lab Order, Lab Test, Billing, and Pharmacy Walk-in Sale.
- Patient records are linked to visits, pharmacy requests, laboratory orders, and billing records.
- User records identify the staff responsible for patient admission, requests, laboratory orders, check-ins, and walk-in sales.
- Primary and foreign keys connect related records across the system



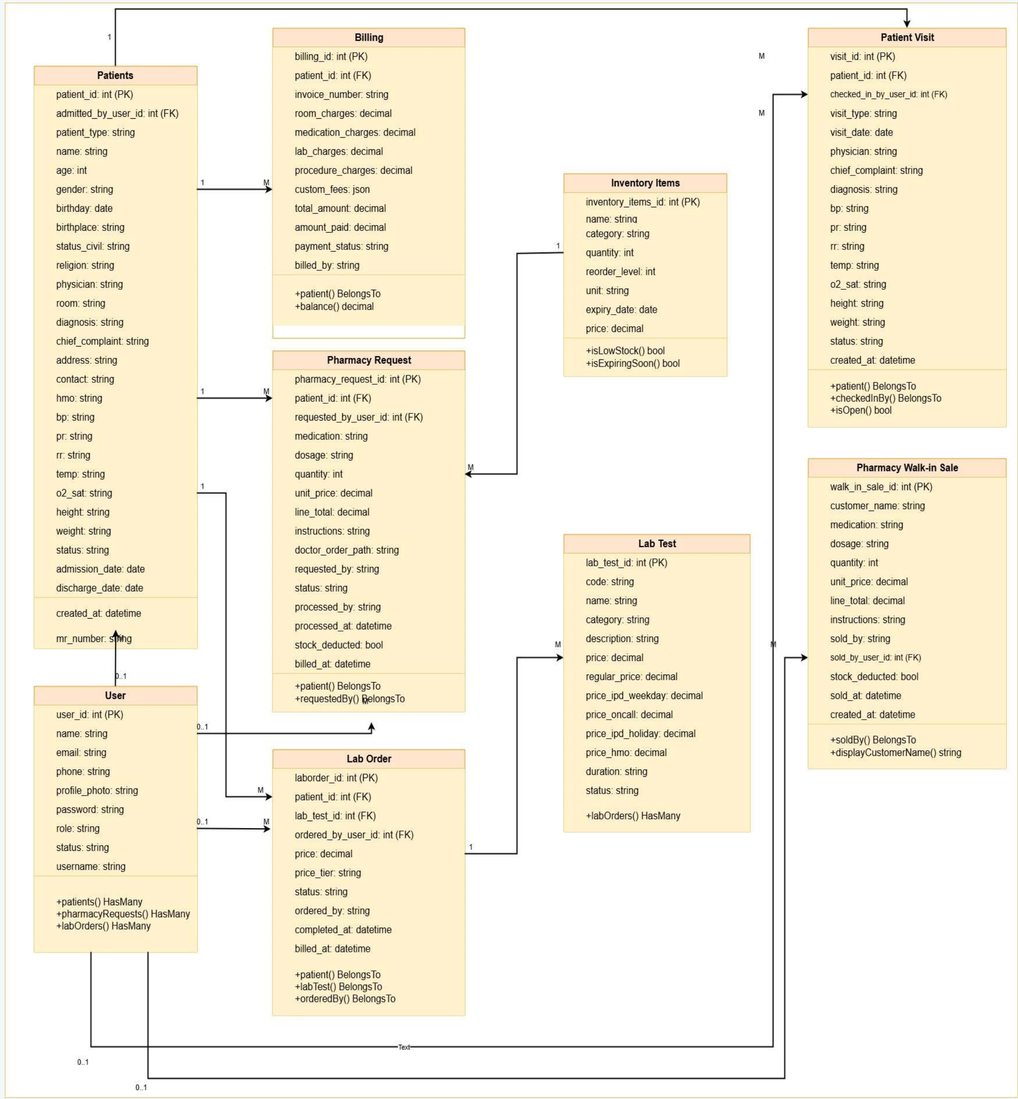
USE CASE (DIAGRAM)

The **Admin** manages user accounts, patient records, inventory, laboratory tests, billing, reports, and predictive decision support.
The **Nurse** handles patient admission, patient updates and discharge, pharmacy requests, and laboratory orders through the mobile portal.
The **Pharmacy Staff** processes pharmacy requests, manages inventory, handles counter sales, and accesses the AI chatbot.
The **Billing Staff** views patient records and laboratory prices, manages payments, and prints or downloads the Statement of Account



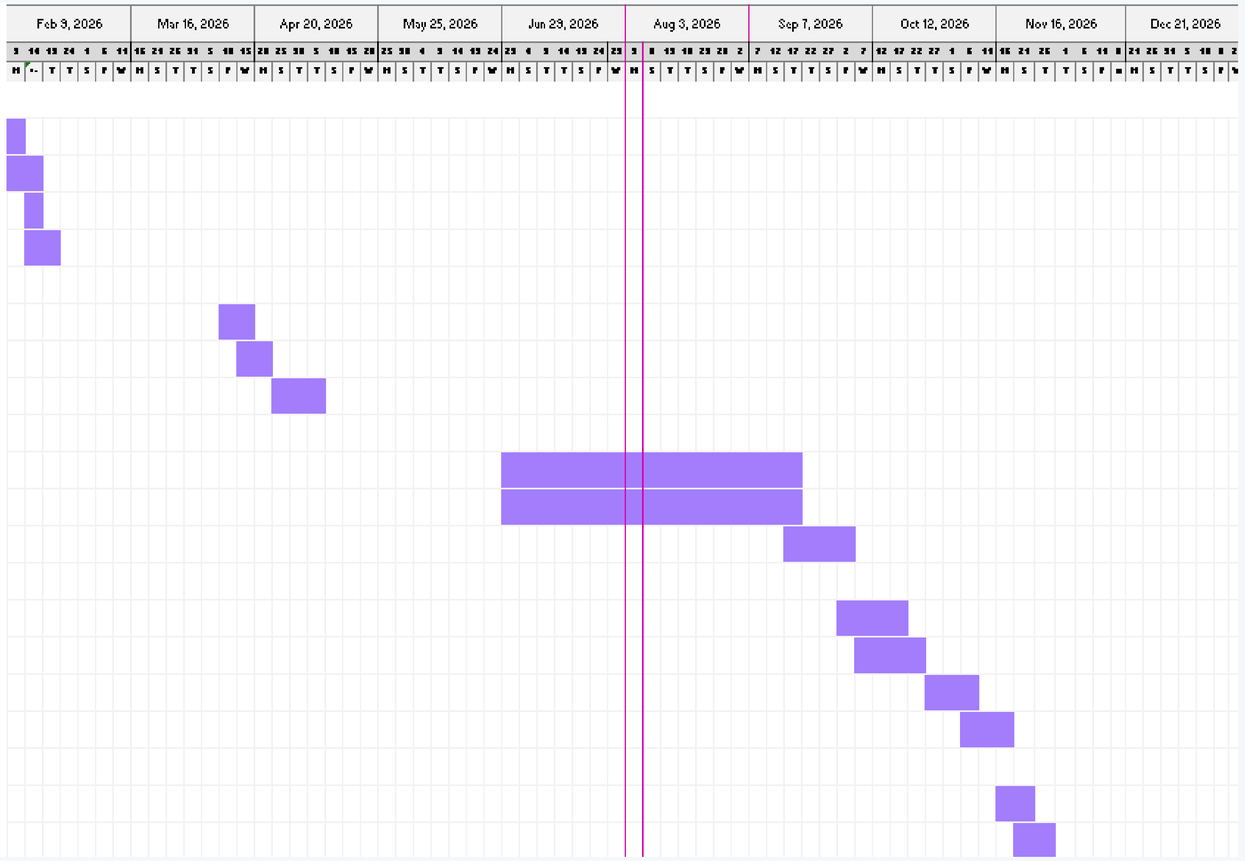
CLASS DIAGRAM (DIAGRAM)

- Each class contains attributes with defined data types, such as patient information, medicine quantities, laboratory prices, and payment amounts.
- Methods define functions such as calculating the billing balance, checking low stock and expiring medicines, and identifying open patient visits.
- Relationships such as **HasMany** and **BelongsTo** connect users, patients, pharmacy requests, laboratory orders, and other records.
- These classes support the organization of patient information and hospital transactions within the system.



GANTT CHART

TASK	ASSIGNED TO	START	END
Requirement Analysis			
Define Problem	Rayner Dely	2/3/26	2/12/26
Define Scope	Rayner Dely	2/12/26	2/15/26
Define Expected Output	Heart Necesito	2/15/26	2/17/26
Define Objectives	Mark Angelo Lazaro	2/17/26	2/20/26
System Design			
Modeling Diagrams	All	4/11/26	4/16/26
Hardware/Software Specification	Heart Necesito	4/17/26	4/24/26
UI Design	Rayner Dely & Mark Angelo Lazaro	4/25/26	5/5/26
Development			
Frontend Design	All	7/1/26	9/13/26
Backend Design	All	7/1/26	9/13/26
Database Design	All	9/20/26	10/5/26
Testing			
Unit Testing	All	10/6/26	10/21/26
Integration Testing	All	10/11/26	10/26/26
System Testing	All	10/27/26	11/6/26
User Acceptance Testing (UAT)	All	11/7/26	11/17/26
Deplogment			
Production Deployment	All	11/18/26	11/24/26
User Training	All	11/25/26	11/30/26



- Shows the timeline and duration of the project from February 2026 to November 2026.
- Illustrates the sequence of tasks, from initial requirement analysis through deployment and evaluation.

CONCLUSION

This project aims to improve hospital operations by connecting the Admin, Nursing, Pharmacy, and Billing departments through an Integrated Hospital Information System that reduces paperwork, improves coordination, and supports accurate and timely billing.

Open for Questions and Suggestions

THANK YOU!

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OUTLINE DEFENDED!

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